

Bluetooth Control Lab · Premium Course

2-session series (2 hrs each) · 11–15 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Control an Arduino gadget wirelessly from a phone over Bluetooth, then build a custom app to command it.

DETAIL

Track	Arduino Robotics & Electronics
Format	2-session series (2 hrs each)
Ages	11–15 yrs
Level	Intermediate → Advanced
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Kits are lent during sessions; buy one to keep building at home.
Parts pack	Optional, KES 1,000: HC-05 Bluetooth module, Small DC motor
Standalone	No
Prerequisites	Button controlled LED (free, recommended)

Programme

1. **Connect & command.** Wire an HC-05 Bluetooth module, pair it with a phone, and send commands from a terminal app to switch LEDs and drive a motor — understanding the serial link underneath.

2. **Build your app.** Use MIT App Inventor to make a custom phone app with your own buttons, and send live sensor readings (distance or temperature) back to the phone for true two-way communication.

What they take home

 Bluetooth-controlled gadget + their own phone app

Why parents love it

“Their phone becomes the remote control — they wire the hardware, build the app, and make the two talk to each other.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Bluetooth Control Lab**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.